

Dario Coscia

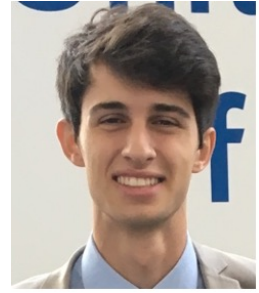
Nationality: Italian

Place and date of birth: Rome, 02 September 1999

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in <https://www.linkedin.com/in/dario-coscia/>

github <https://github.com/dario-coscia>



Work Experience

2022 – 2023

Machine learning researcher

Research fellow at SISSA *mathLab* group in the field of Deep Learning.

- Working uncertainty quantification in scientific machine learning; particular focus on fluid dynamics and simulation-based chemistry.
- Developing Physics Informed Neural Networks for Advanced modelling software ([PINA](#)). The software is currently used by many universities, with 10+ maintainers from diverse institutions around the world

2020 – 2021

CNR-IOM

Internship at CNR-IOM Trieste in the field of computational solid-state physics. Building statistical and energetic models for material design.

Education

2023 –

PhD Student

SISSA, Trieste Italy - UvA, Amsterdam - The Netherlands

Bayesian and Generative Modelling, Uncertainty Quantification for Material Science and PDE modelling.

2021 – 2023

Master's degree Data Science and Scientific Computing

University of Trieste, Trieste Italy

Artificial Intelligence and Machine learning path

Thesis title: *A generative adversarial method for reduced order modelling*

2018 – 2021

Bachelor's degree Physics

University of Trieste, Trieste Italy

Grade: 110/110 “cum laude” (graduated with distinction)

Thesis title: *Modelling the energetics of graphene on a Nickel surface*

2016 – 2018

United World College of South East Asia

Singapore, High School

International Baccalaureate (English language)

Grade: 40/45 and bilingual diploma

Research Publications

1

Coscia, D., Welling, M., Demo, N., & Rozza, G. (2025). BARNN: A Bayesian Autoregressive and Recurrent Neural Network. *International Conference of Machine Learning*.

- 2 Coscia, D., Demo, N., & Rozza, G. (2024). Generative Adversarial Reduced Order Modelling. *Nature Scientific Reports*, 14(1), 3826.
- 3 Coscia, D., Ivagnes, A., Demo, N., & Rozza, G. (2023). Physics-Informed Neural networks for Advanced modeling. *Journal of Open Source Software*, 8(87), 5352.
- 4 Coscia, D., Meneghetti, L., Demo, N., Stabile, G., & Rozza, G. (2023). A Continuous Convolutional Trainable Filter for Modelling Unstructured Data. *Computational Mechanics*, 1–13.

Invited Talks

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|---------------|---|---|
| July 23, 2025 | ■ | Brown University (U.S.A.), CRUNCH Webinar: <i>A Variational Bayesian method for Uncertainty Quantification in Autoregressive and Recurrent Neural Networks</i> - seminar . |
| July 15, 2025 | ■ | International Centre for Theoretical Physics (Italy), Advanced School on Foundation Models for Scientific Discovery: <i>Uncertainty Quantification in Scientific Machine Learning - Theory and Recent Advancements</i> - summer school . |
| June 26, 2025 | ■ | Banff International Research Station (Canada), Efficient and Reliable Deep Learning Methods and their Scientific Applications: <i>A Variational Bayesian Method for Sequence Models Predictions and Uncertainty Quantification</i> - seminar . |
| June 19, 2025 | ■ | KTH Royal Institute of Technology (Sweden), PINN Summer School: <i>PINA, a Python Software for Scientific Machine Learning</i> - summer school . |
| June 18, 2025 | ■ | KTH Royal Institute of Technology (Sweden), PINN Summer School: <i>Introduction to Uncertainty Quantification in Scientific Machine Learning</i> - summer school . |
| May 12, 2025 | ■ | International Centre for Theoretical Physics (Italy), Summer School on Theory, Mechanisms and Hierarchical Modelling of Climate Dynamics: <i>Round Table Discussion on Uncertainty Evaluation for Climate Modelling</i> - summer school . |
| Apr 29, 2025 | ■ | Heidelberg University (Germany), Institute for Theoretical Physics seminars: <i>A Variational Bayesian Method for Sequence Models</i> . |
| Feb 11, 2025 | ■ | University of Padova (Italy), GNCS-SIAM Chapters Meeting for Young Researchers in Numerical Analysis and Applied Mathematics: <i>Bayesian Autoregressive and Recurrent Neural Networks for Neural PDE Solvers Uncertainty Quantification and Drug Discovery</i> - conference . |
| May 15, 2024 | ■ | Electrolux SPA: <i>Physics Informed Machine Learning for Accelerating Numerical Simulations</i> - seminar . |
| Jan 24, 2024 | ■ | Area Science Park (Italy), LADE seminars: <i>Inductive Bias in Scientific Machine Learning for Modelling PDEs</i> - seminar . |
| Dec 19, 2023 | ■ | University of Trieste (Italy), iNEST spoke: <i>Deep Learning for Solving Differential Equations</i> - seminar . |

Awards and Achievements

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|------|---|---|
| 2023 | ■ | Best thesis award , for the best thesis on Artificial Intelligence at the University of Trieste. |
| 2021 | ■ | SISSA merit scholarship , the MSc in Data Science and Scientific Computing at University of Trieste SISSA ICTP and University of Udine |
| | ■ | Collegio Universitario Luciano Fonda Scholarship , for outstanding students studying at the University of Trieste |
| 2019 | ■ | Lucia Malagnini Physics Scholarship , for the best student in the Physics department at the University of Trieste |

Awards and Achievements (continued)

2016  **United World College Scholarship**, for outstanding students studying in Italy to attend the prestigious UWC college.

Additional information

Languages  **Italian:** native speaker
English: full professional
Spanish: professional working

Computer skills  High experience in Python programming language

- High experience in PyTorch, PyTorchLightning framework for Deep Learning
- Experience in ML software: Scikit-learn, Pyro and JAX

Experience in C++ and Fortran 90 programming languages
Experience in parallel computing, distributed training and software development

Reference Available Upon Request

Curriculum Vitae last update: August 10, 2025